

MIDAS GUI — User Documentation

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Version: 1.0.0 **Application:** `midas-gui` (or `python -m midas_gui`) **Backends:** `midas_calibrate_v2`, `midas_integrate_v2`, `midas_calibrate`, `midas_hkls`, `midas_distortion` **Last updated:** 2026-07-06

Maintenance: keep this document in sync with the code — whenever the workflow or a tab's controls change, update the relevant section here in the same change.

Tab status: Tabs **0–4** (Data Viewer, Mask Builder, Calibrate, Calib. Refinement, Batch Integrate) are **verified** and ready to use. Tabs **5–8** (Corrections & Physics, PDF Analysis, Texture, Results & Export) are a **work in progress** and will be updated in the coming weeks.

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1. Overview and Architecture

The MIDAS GUI is a 9-tab PyQt5 desktop application that exposes the scientific capability of the `midas_calibrate_v2` and `midas_integrate_v2` packages through a structured workflow. The intended order of use is:

```
Data Viewer (inspect) → Mask Builder → Calibrate → [Refine] → Batch Integrate
                                     → Corrections preview
                                     → PDF Analysis
                                     → Texture / Pole Figure
                                     → Results & Export
```

Cross-tab shared state. When Tab 2 (Calibrate) produces a result it is automatically propagated to all downstream tabs. When Tab 1 (Mask Builder) computes a mask it is sent to the consuming tabs. Tab 3 (Refinement), if applied, re-broadcasts the refined geometry. No manual copying is needed.

All heavy computation runs off the GUI thread. Every operation that touches a MIDAS package (calibration, integration, mask training, PDF transform, gain training, field averaging, drift fitting) runs in a background `QThread` worker; the GUI stays responsive and log lines stream in real time.

Package layout.

```
midas_gui/
├─ app.py           MainWindow, dark theme, crash diagnostics, main()
├─ __main__.py      enables `python -m midas_gui`
├─ style.py         Dioplas-inspired QSS theme + layout helpers
├─ constants.py     calibrants, colormaps, dtype sentinels, default paths
├─ helpers.py       image IO, geometry parsing, spec building, no-scroll widgets
├─ widgets.py       ImageViewer / PickableImageViewer / ProfileViewer / FieldSelector / ...
├─ workers.py       all QThread workers + the integration core
├─ calib.py         calibration-pipeline dispatch
├─ dialogs.py       save dialogs
└─ tab_*.py         one module per tab
```

Layout pattern. Every tab has a fixed-width scrollable left control panel and a right display area (image viewer, plots, and/or a log panel).

2. Getting Started

`midas-gui` is **not on PyPI** — install it from source with conda.

```
git clone https://github.com/d-beniwal/MIDAS_GUI.git
cd MIDAS_GUI
conda env create -f environment.yml    # creates the 'midas-gui' env and installs the GUI
conda activate midas-gui
midas-gui                             # launch (equivalently: python -m midas_gui)
```

The GUI drives the MIDAS analysis backends (`midas-calibrate-v2`, `midas-integrate-v2`, `midas-calibrate`, `midas-hkls`, `midas-distortion`), which are also not on PyPI — point the `pip:` section of `environment.yml` at your MIDAS source before creating the environment (see the comments in that file).

Test data

Synthetic test data lives in the repo-root `test_data/` folder (git-ignored): - `calibrant_ceria.tif` / `.h5` — CeO₂ calibrant (Eiger2 500K, 1028×512, 75 µm pixels) - `nickel_tifs/` — 10-frame Ni scan, lattice expanding +0.1 %/frame - `nickel_stack.h5` — the same scan as one HDF5 stack (`exchange/data`)

Default geometry: $\lambda = 0.39$ Å, Lsd = 121 mm, pixel = 75 µm, BC = (10, 10) px. Every tab's default file paths point at this data and **auto-load on startup when present**, so the app is usable immediately after checkout.

3. Tab 0 — Data Viewer

Purpose: inspect detector frames and do a quick geometry-free radial integration. Produces no shared state for other tabs.

Data card

Field	Description
Path field	File, folder, or glob (e.g. <code>frames/*.tif</code>). Selecting via browse — or pressing Enter — loads immediately (no separate Load button).
<code>...</code> button	Browse for a file.
Browse folder...	Browse for a folder of frames (loads immediately).
Dataset (HDF5)	Auto-populated dropdown of every ≥ 2 -D dataset with its shape; changing it reloads.

Frame navigator

Appears for a multi-frame stack — slider, spin box, or ◀/▶. **Zoom/pan is preserved as you step through frames** (the view only auto-frames on a fresh load).

Projection card

Collapse a stack to one image: **Max** (hot-pixel hunting), **Sum** (long-exposure equivalent), or **Average** (noise reduction), along a chosen axis (0 = across frames).

Calibration card (optional)

Load geometry from a **calibration .json**, a **MIDAS paramtest.txt**, or a **pyFAI .poni** (auto-detected). It fills BC, Lsd, pixel size and wavelength, unchecks “Beam centre = image centre”, and refreshes the overlay and radial plot.

Intensity range card (radial integration)

Field	Description
Exclude out-of-range pixels	When on, pixels $\leq$ min or $>$ max are drawn as a red overlay and excluded from the radial integration (removes gaps / hot / overflow).
pixel $\leq$ / pixel $>$	Lower / upper bounds. On load, the upper bound auto-fills to max(99.99th percentile, 100000) .

Ring simulation card

Overlays simulated Debye-Scherrer rings to check geometry. - **Material** dropdown (CeO₂, LaB₆, Si, Al₂O₃, Cu, Ni, FCC-γFe, BCC-αFe, Au, Ag, Pt, W, Ti) or **Custom**. - Lattice entry is compact: **a**, **b**, **c** on one row and **α**, **β**, **γ** on the next, plus **SG #**. A **Cubic (a=b=c, α=β=γ=90°)** checkbox lets you enter only **a** for cubic crystals (b, c mirror a; angles fixed at 90°). Lattice fields are editable only for Custom. - Geometry: λ, max 2θ, Lsd, pixel size, and beam centre (auto = image centre, or manual BC_y/BC_z). - **Show rings / Labels** toggles; **Simulate rings** draws the overlay + hkl table.

Beam-centre picking (on the image)

The image viewer has **Pick BC** (single click sets the beam centre) and **Pick Ring** (click ≥ 3 points on a ring; a circle fit estimates the beam centre). Either updates BC_y/BC_z and re-runs the overlay + radial integration.

Radial integration plot (bottom-right)

Below the image is a live **azimuthal average about the beam centre** (**R bin** , **Integrate** , and an **Auto** toggle that recomputes on frame/BC/mask change). It is a simple geometry-free radial average (flat detector, no tilt/distortion). **Clicking a radius on the plot draws the matching ring (magenta) on the image**. Axis units switch between R (px) / 2θ / Q.

4. Tab 1 — Mask Builder

Purpose: identify and exclude bad pixels. The final mask is a uint8 array (1 = bad) broadcast to Tabs 2, 3, 4, 5. Browsing an image or a mask loads it immediately.

Section 1 · Threshold mask (always applied)

pixel $\leq$ lower | pixel $>$ upper . The upper bound auto-fills from the data type on load (e.g. 1,048,575 for uint20 Eiger, 4,294,967,295 for uint32).

Section 2 · Statistical auto-mask

Robust spatial-outlier detection (5×5 median residual → 15×15 local MAD → Z-score → hot/dead/saturated gates). Uses a **temporal median** over a stack folder when provided (much cleaner reference). Controls: **K_σ** (6.0), **Hot** (1.5), **Dead** (0.5), **Frozen** temporal-constancy fraction (0.05, needs ≥ 2 frames), stack folder + stride.

Section 3 · Spatial spike rejection

Laplacian high-pass single-pixel-spike detector; `n_σ` threshold (5.0).

Section 3b · Cosmic-ray rejection (temporal)

Per-pixel temporal σ -clip across a ≥ 3 -frame stack; anomalies OR'd across frames. `n_σ` (5.0).

Section 4 · Calibration-based masks (need a Tab 2 result)

- **Azimuthal σ -clip** — flags (R, η) cells deviating from the azimuthal mean.
- **Learnable mask** — differentiable per-pixel mask trained against an η -uniformity loss (`steps` , `lr` , `sparsity`).

Draw mask tools

Interactive rectangle / oval / circle / polygon / annulus / point tools, live-draggable; **Apply shapes** → **mask** rasterises them (OR'd with the computed mask). **Clear shapes** removes them.

Save / Load

Save the combined mask as TIFF (0 = good, 1 = bad); loading a TIFF applies immediately.

5. Tab 2 — Calibrate

Purpose: determine detector geometry (Lsd, BC, tilts, distortion, wavelength) from a calibrant pattern.

Files

Load a TIFF/HDF5 calibrant (browse or Enter loads immediately; the HDF5 dataset field and Frame index reload on change). A mask file can be loaded here too.

Dark / Bright / Background

A card with three **field selectors**. Each builds one 2-D field from a **single file, a folder/glob, or an HDF5 dataset** (with a container dropdown), averaged over a chosen **index range** (clamped to the frames available). The group is compact — each field's body is hidden until its checkbox is ticked. The **Bright** field has a mode: **Flat-field divide** or **Subtract**. Correction order: `(img - dark)` → bright divide/subtract → `- background` → clip ≥ 0 .

Pipeline selector

One-shot (default) · First-time · Four-stage · Bayesian (Laplace σ) · Joint-cake. *For trustworthy tilt/strain, prefer Four-stage or First-time — One-shot/Bayesian can report a spurious self-compensated tilt on weakly-tilted data.*

Refine flags

Which parameters vary: Lsd, BC, ty, tz, tx, Wavelength, Distortion (15 coeffs), plus a “Residual map” build toggle. Advanced (E-M / LM iters, device, output dir) and Multi-panel detector groups are collapsible.

Live threshold slider

Zeroes calibration-image pixels below the slider value (background suppression for ring-finding); the preview updates instantly.

Pick BC / Pick Ring

Click the image to seed the beam centre (single click) or fit a ring (≥ 3 clicks).

Run / Abort

Run Calibration launches the worker; **Abort** terminates it and frees the slot so you can immediately start a new run (the calibration is one uninterruptible library call, so abort hard-terminates the worker thread rather than waiting).

Results (right panel, bottom tabs)

Radial Profile (with ring markers and an **Azim. avg** selector — see Tab 4), Ring Residuals bar chart, Results (Lsd/BC/strain + distortion table), and Log. Integration runs automatically after calibration.

Export

Save calibration.json and **Save paramstest.txt** (standalone or from a template).

6. Tab 3 — Calibration Refinement

Purpose: optional post-calibration geometry refinement against an **η -uniformity** criterion (rings should be azimuthally uniform). Uses a **derivative-free Nelder-Mead** optimiser (the differentiable integrator returns NaN geometry gradients at the beam-centre singularity on this build).

Controls: calibration source (Tab 2 result or JSON), sample frame (auto-loads on browse), parameters to refine (BC_y, BC_z, Lsd, ty, tz), max iterations (500), tolerance ($1e-6$). **Run Refinement** shows a live loss curve; **Apply** broadcasts the refined geometry downstream. If Apply is never clicked, the Tab 2 result flows through unchanged.

7. Tab 4 — Batch Integrate

Purpose: integrate a stack of frames into 1-D profiles using the calibrated geometry and optional corrections.

Calibration source

- **From Tab 2** — the calibration (or refined) result.
- **From file** — a **calibration .json**, **MIDAS paramstest.txt**, or **pyFAI .poni** (auto-detected; both GUI and MIDAS-pipeline json key styles supported). Entering a path auto-selects this option.

Data files

TIFF folder/glob or HDF5 (with dataset). **Streaming controls:** start / end (0 = all) / stride.

Dark / Bright / Background

Same three field selectors as the Calibrate tab (file/folder/HDF5, index-range average, bright divide or subtract), applied per frame.

Mask

Load a mask TIFF (browse loads immediately) or use the Tab 1 mask.

Integration

Field	Description
Kernel	Hard (fastest) · Subpixel K=2 · Subpixel K=4 · Polygon (exact).
R bin (px) / η bin (°)	Radial and azimuthal bin sizes.
Azim. avg	How the (η , R) cake becomes a 1-D profile: Pixel-weighted (default) $\Sigma(\text{mean} \cdot \text{count}) / \Sigma(\text{count})$ — independent of η -bin size and robust to partial azimuthal coverage / off-detector beam centres ; or η-bin mean (legacy) — the unweighted mean of per- η -bin means, which can distort with a coarse η bin when the beam centre is off the detector.
Per-bin variance (σ)	Error model poisson / azimuthal / hybrid (ignored when corrections are on $\rightarrow \sigma = \sqrt{I}$).
Q-uniform bins	Integrate in R then rebin onto a uniform-Q grid (Qmin, Qmax, ΔQ).

Physics corrections

Polarization and solid-angle (pixel-domain, via `integrate_with_corrections`).

Monitor normalisation

Divide each processed frame's profile/ σ by a per-frame scalar from a text file (one value per *processed* frame).

Drift correction (long scans)

Per-frame geometry interpolated from an anchor JSON (`{frame_idx: {Lsd, BC_y, BC_z}}`) with spline/linear/constant parametrization; **Fit trajectory** builds it. Each frame is then integrated with its own geometry.

Run / Abort

Start Integration / Abort. Abort first asks the worker to stop cleanly between frames (keeping frames already written); if it does not stop promptly it force-terminates and frees the slot. Completed frames' output files are preserved.

Output formats

CSV (R,I, σ) · XYE (2 θ) · FXYE (centideg) · DAT (Q) · HDF5 (full stack) · 2D-CSV ($\eta \times R$ cake). Right panel: live **Waterfall** and **Stacked profiles**.

8. Tab 5 — Corrections & Physics (*work in progress*)

Preview physics corrections on a single frame (auto-loads on browse). Pixel-domain: polarization, solid angle, empty subtraction. Profile-domain: cylindrical absorption (μR), Compton subtraction (composition:fraction). **Compute corrected profile** overlays corrected vs uncorrected and shows the correction factor vs 2 θ .

Also hosts **per-pixel gain training (LearnableGain)**: from a clean reference frame and a drifted frame, learn a spatial gain map $g_i = 1 + \text{scale} \cdot r_i$ by minimising $\text{MSE}(\text{profile}) + \text{unity} \cdot \sum (g-1)^2 + \text{smooth} \cdot \text{TV}(g)$; save as NPZ and apply with `corrected = raw / gain_map`.

9. Tab 6 — PDF Analysis (*work in progress*)

Compute G(r) from one calibrated frame: image $\rightarrow I(Q) + \text{background} \rightarrow F(Q) \rightarrow G(r)$. Controls: Qmin/Qmax/ ΔQ , rmin/rmax/ Δr , window (Lorch/none), binning (hard/polygon). Right panel: $I(Q) + \text{background}$, $F(Q) = Q \cdot (I/bg - 1)$, and G(r). **Save G(r)** writes a three-column `r, G(r),  $\sigma$` file (diffpy-CMI / PDFgui / RMCProfile compatible).

10. Tab 7 — Texture / Pole Figure (*work in progress*)

Per-ring azimuthal analysis for preferred orientation. Controls: calibration source, sample frame, R/ η bins, ring index, χ (tilt) / ϕ (rotation). **Compute Pole Figure** integrates to a cake and extracts $I(\eta)$ at the selected ring; the right panel shows the stereographic pole figure and the raw $I(\eta)$. **Save pole figure (.pol)** exports POPLA format.

11. Tab 8 — Results & Export (*work in progress*)

Session summary + one-click export. Checkboxes select which products (calibration.json, paramstest.txt, mask.tif, integrated profiles, G(r), pole figures, session log) to copy to an output directory. A provenance block (package versions, geometry hash, mask fraction, correction flags) can be copied to the clipboard for a Methods section.

12. Common UI Conventions

- **Browse = load.** Selecting a file/folder (or pressing Enter in a path field) loads it immediately; there are no separate “Load” buttons. HDF5 dataset / frame-index changes reload automatically.
 - **No accidental scroll changes.** Spin boxes and drop-downs ignore the mouse wheel — values change only by clicking/typing; the wheel scrolls the panel instead.
 - **Readable right-click menus.** pyqtgraph plot context menus use the dark theme.
 - **Dark theme, orange accent** (Dioptas-inspired); off-white text, light input fields.
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13. Packaging, Deployment & Diagnostics

Packaging. `midas-gui` is a MIDAS-style package: `pyproject.toml` (BSD-3-Clause), a `tests/` smoke suite, and `release.sh` for cutting versioned releases (see `RELEASING.md`). Once published it can join the `midas-suite` meta-package as an optional `gui` extra (`pip install midas-suite[gui]`).

Deployment on another system. Clone the repo, create the conda environment (`environment.yml`), and run `midas-gui`. The MIDAS analysis backends are installed via the `pip:` section of `environment.yml` (editable from a local MIDAS checkout, from a git subdirectory, or from a private index). `midas_gui/_paths.py` is an inert stub in an installed package (no `sys.path` manipulation).

Crash diagnostics. On startup the app installs a logging exception hook and `faulthandler`; any uncaught Python exception or native fault is written to `~/midas_gui_error.log` and shown in a dialog. Installing the hook also prevents PyQt5 from hard-aborting on an exception inside a slot. Each tab is built in isolation — a tab that fails becomes an error placeholder instead of taking the window down. If the app ever “pops up and dies” (typically on Windows), send that log file.

For bugs or questions, see the MIDAS GUI repository.